Workout 1A Homework 2 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 59,994 of 99,990 points. 60%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1

If $\frac{8}{9}+\frac{10}{11}=1+\cfrac{1}{1+\cfrac{1}{a+\cfrac{1}{b+\cfrac{1}{c}}}}$, where a, b, and c are integers, what are a, b, and c?

a = 1.

b = 2.

c = 3.

Your Answer

1. [no answer given]
2. [no answer given]
3. [no answer given]

Correct Answer

1. 3
2. 1
3. 19

Explanation

First combine the two fractions into one fraction, and then peel off successive 1's from the fraction to create a nested fraction.

$$\begin{aligned}
\frac{8}{9}+\frac{10}{11} &= \frac{8(11)+10(9)}{9\times 11} \\
&= \frac{178}{99} \\
&= 1+\frac{79}{99} \\
&= 1+\frac{1}{\frac{99}{79}} \\
&= 1+\frac{1}{1+\frac{20}{79}} \\
&= 1+\frac{1}{1+\frac{1}{\frac{79}{20}}} \\
&= 1+\frac{1}{1+\frac{1}{3+\frac{19}{20}}} \\
&= 1+\frac{1}{1+\frac{1}{3+\frac{1}{\frac{20}{19}}}} \\
&= 1+\frac{1}{1+\frac{1}{3+\frac{1}{1+\frac{1}{\frac{1}{19}}}}}.
\end{aligned}$$

Thus $a=\boxed{3,}$ $b=\boxed{1,}$ $c=\boxed{19}$.

2 Which of the following is the formula for the sum of the first n squares, starting with $n = 1$?

Your Answer $\frac{n(n+1)(2n+1)}{6}$

Correct Answer $\frac{n(n+1)(2n+1)}{6}$

Explanation

When $n = 1$, the formula should give 1, so this eliminates all except for the first and last choices. When $n = 2$, the sum of the first 2 squares equals $1 + 4 = 5$, so this eliminates the first choice. By process of elimination, the correct answer is $\boxed{\frac{n(n+1)(2n+1)}{6}}$.

3 A recipe for puff pastry calls for rolling out the dough and folding it, done 6 times. How many layers are there in the puff pastry?

Your Answer	[no answer given]
Correct Answer	64

Explanation

Every time the pastry is folded, the number of layers doubles. Thus after 6 folds, there will be $2^6 = \boxed{64}$ layers of pastry.

4 What is 23564541 divided by 1001?

✓	**Your Answer**	23541
	Correct Answer	23541

Explanation

We could do a straightforward division, but it might be more fun to find some clever simplifications. Let n be an integer such that $1001n = 23564541$. Multiplying by 1001 is the same as shifting the digits of a number to the left by three places, and then adding it to itself. The resulting sum is three digits longer than the original number. Since 23564541 is 8 digits long, the original number is 5 digits long. Supposing its digits are $ABCDE$, the sum will be like so:

$$\begin{array}{r} ABCDE \\ ABCDE \\ \hline 2\,3\,5\,6\,4\,5\,4\,1 \end{array}$$

The first 2 digits and the last 3 digits of 23564541 can be read off as A, B, C, D, and E, since they are unaffected by carries. Thus $A = 2, B = 3, C = 4, D = 4$, and $E = 1$. The original number is then $\boxed{23541}$.

5 Given that

$$(a+4)(a+3)(a-2)-(a+4)(a-6)=1400,$$

for integer a, find the value of a.

✓	**Your Answer**	10
	Correct Answer	10

Explanation

Factor out common terms, like $(a+4)$. We get:

$$\begin{aligned}(a+4)\Big((a+3)(a-2)-(a-6)\Big)&=1400\\(a+4)\Big(a^2-2a+3a-6-a+6\Big)&=1400\\(a+4)\left(a^2\right)&=1400\end{aligned}$$

It's fairly easy to see that $a=\boxed{10}$, and plugging this in, we get $a^2(a+4)=10^2(10+4)=1400$ ✓.

6 Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ each to the winner of the game. If Arnav ends up winning a net $\$15$, how many games did he win?

✓	**Your Answer**	15
	Correct Answer	15

Explanation

Let a equal the number of games Arnav won. Then $30 - a$ equals the number of games he lost. He wins $\$2$ from each win and loses $\$1$ from each loss, so his net money equals $\$2a - (30 - a)$. Setting this equal to the $\$15$ he nets in total, we get $3a - 30 = 15$, so $3a = 45$ and $a = \boxed{15}$ games.

7 Without using a calculator, what are the first 3 digits of $\frac{123456789}{1000001}$?

Your Answer	[no answer given]
Correct Answer	123

Explanation

We know the value of $\frac{123456789}{1000000}$ is 123.456789. If we take the difference between these two fractions, we get

$$\frac{123456789}{1000000} - \frac{123456789}{1000001}$$

$$= \frac{123456789 \times 1000001 - 123456789 \times 1000000}{1000000 \times 1000001}$$

$$= \frac{123456789}{1000001000000}$$

This is much smaller than 1. Since the tenths digit of $\frac{123456789}{1000000}$ is 4, there is no way this difference will have any affect over the first 3 digits. Therefore, the first 3 digits will still be $\boxed{123}$ in our original fraction.

8 Suppose

$$\begin{aligned} a+c+d &= -14 \\ b+c+d &= 6 \\ a+b+d &= -9 \\ a+b+c &= 5. \end{aligned}$$

What is the value of b?

✓	**Your Answer**	10
	Correct Answer	10

Explanation

Adding the four equations gives us $3(a+b+c+d) = -14+6-9+5 = -12$. Dividing by 3, we find that $a+b+c+d = -4$. We can use the first equation, $a+c+d = -14$, to solve for b.

$$\begin{aligned} a+b+c+d &= -4 \\ b+(-14) &= -4 \\ b &= \boxed{10}. \end{aligned}$$

9 Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Arnav ends up winning a net $\$15$ and Bing loses $\$9$ more than Chirag, then how many games did Chirag win?

✕	**Your Answer**	[no answer given]
	Correct Answer	9

Explanation

Let a equal the number of games win by Arnav, b equal the number of games won by Bing, and c equal the number of games won by Chirag. For Arnav, we saw earlier that he earns $2a$ from his won games and loses $30-a$ dollars from his lost games, giving him a net $2a-(30-a)=3a-30$ dollars. We can write similar equations for Bing and Chirag.

$$\begin{aligned}\text{Arnav's net money} &= 3a-30\\ \text{Bing's net money} &= 3b-30\\ \text{Chirag's net money} &= 3c-30\end{aligned}$$

Bing loses $\$9$ more than Chirag, so Chirag won more games. The difference in their winnings can be written as:

$$\begin{aligned}\text{Chirag's winnings} - \text{ Bing's winnings} &= 3c-30-(3b-30)\\ 9 &= 3c-3b\\ 9 &= 3(c-b)\\ 3 &= c-b\end{aligned}$$

This tells us that Chirag won 3 more games than Bing. We found from a previous question that Arnav won 15 games, since $3a-30=15 \implies a=15$. There are 15 games won by either Chirag or Bing, so we can write:

$$\begin{aligned}15 \text{ games } &= (c-3)+c\\ 15 &= 2c-3\\ 18 &= 2c\\ 9 &= c.\end{aligned}$$

Thus Chirag won $\boxed{9}$ games.

10 What is the sum $15+17+19+21+23+25+27+29+31+33+35$?

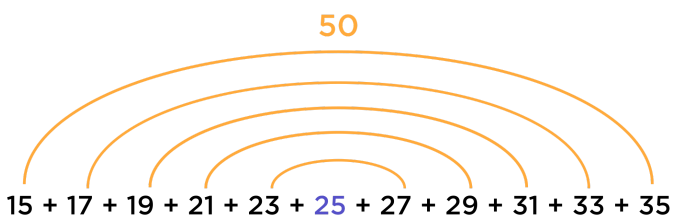
✓	**Your Answer**	275
	Correct Answer	275

Explanation

Solution 1. Use the formula for the sum of the first n odds. We saw in the Week 1 Challenge that the sum of the first n odds equals n^2. We can write the nth term as $1+2(n-1)$, so $35=2+2(18-1)$ is the 18th odd. Thus $1+3+5+7+\ldots+33+35=18^2=324$. We must subtract $1+3+5+7+9+11+13$, which is the first 7 odds, which sums to $7^2=49$. Subtracting this from 324, we get $\boxed{275}$.

Solution 2. Use the average.

$$\text{sum } = (\text{average}) \times (\text{number of numbers})$$



We can pair up the numbers to create sums equal to 50. There are 5 pairs, with a 25 left over. The total sum is then $50\times 5+25=250+25=\boxed{275}$.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Arnav ends up winning $\$6$ more than Bing and Bing wins $\$6$ more than Chirag, then how many games did Bing win?

Your Answer	[no answer given]
Correct Answer	10

Explanation

We can use symmetry as a shortcut in this problem! Bing is the average money winner, since Arnav wins $\$6$ more and Chirag wins $\$6$ less than Bing. Thus he wins the average number of games, which is equal to $\frac{30}{3} = \boxed{10}$ games.

12 What is the sum of the following?

$$\begin{gathered}1001+1111+1221+1331+1441+1551+1661+1771+1881+1991+\\2002+2112+2222+2332+2442+2552+2662+2772+2882+2992+\\3003+3113+3223+3333+3443+3553+3663+3773+3883+3993+\\4004+4114+4224+4334+4444+4554+4664+4774+4884+4994+\\\cdots\\9009+9119+9229+9339+9449+9559+9669+9779+9889+9999.\end{gathered}$$

Your answer should the number only, with no commas.

Your Answer	[no answer given]
Correct Answer	495000

Explanation

Pair them up. We have 90 numbers, and we can make 45 pairs where each pair sums to 11000. For example, $1771+9229=11000$ and $1001+9999=11000$. Thus the answer is $11000\times 45=\boxed{495000}$.

$$\begin{gathered}1001+1111+1221+1331+1441+1551+1661+1771+1881+1991+\\2002+2112+2222+2332+2442+2552+2662+2772+2882+2992+\\3003+3113+3223+3333+3443+3553+3663+3773+3883+3993+\\4004+4114+4224+4334+4444+4554+4664+4774+4884+4994+\\\cdots\\9009+9119+9229+9339+9449+9559+9669+9779+9889+9999\end{gathered}$$

13 What is the sum of the digits of the decimal that is equivalent to 5^{-6}?

Your Answer	[no answer given]
Correct Answer	10

Explanation

There is a neat pattern with the digits that make up a power of 5; they relate to the digits of the powers of 2. Use the fact that negative exponents are the reciprocals of their positive counterparts ($x^{-n} = \frac{1}{x^n}$), to write the following:

$$\begin{aligned} 5^{-6} &= \frac{1}{5^6} \\ &= \left(\frac{10^6}{10^6}\right)\frac{1}{5^6} \\ &= \left(\frac{1}{10^6}\right)\frac{10^6}{5^6} \\ &= \left(\frac{1}{10^6}\right)2^6 \end{aligned}$$

The final expression is just 2^6 with its decimal point shifted 6 places to the left. Thus $5^{-6} = \frac{64}{10^6} = 0.000064$, and the sum of the digits of the number is $4 + 6 = \boxed{10}$.

14 The first term of a sequence is equal to $(1007 + 2008 + 3009)$, the second term is equal to $(1014 + 2016 + 3018)$, the third term is equal to $(1021 + 2024 + 3027)$. What is the sum of the first 14 terms of this sequence?

Your Answer	[no answer given]
Correct Answer	86520

Explanation

The numbers in the 1000's happen to be $1000 + (\text{a multiple of } 7)$, the numbers in the 2000's are $2000 + (\text{a multiple of } 8)$, and the numbers in the 3000's are $3000 + (\text{a multiple of } 9)$. Let's use this fact to group the numbers based on what multiple they contain.

$$\begin{aligned}&(1007 + 1014 + 1021 + \ldots + 1000 + 7(14))\\ &+ (2008 + 2016 + 2024 + \ldots + 2000 + 8(14))\\ &+ (3009 + 3018 + 3027 + \ldots + 3000 + 9(14)).\end{aligned}$$

This is already simpler than before, but we can do a little better by separating out the fourteen sets of $1000 + 2000 + 3000$ and then grouping the other numbers by multiple of 1, multiple of 2, multiple of 3, etc., up to multiple of 14.

$$\begin{aligned}\text{sum} &= 14(1000) + 7(1 + 2 + 3 + \ldots + 14) + 14(2000) + 8(1 + 2 + 3 + \ldots + 14) + 14(3000) + 9(1 + 2 + 3 + \ldots + 14)\\ &= 14(1000 + 2000 + 3000) + (7 + 8 + 9)(1 + 2 + 3 + \ldots + 14)\\ &= 14(6000) + 24(1 + 2 + 3 + \ldots + 14)\end{aligned}$$

Here let's employ the trick for adding the terms of a consecutive arithmetic series, which saw in Day 7. Then we have $1 + 2 + 3 + \ldots + 14 = \frac{14 \times 15}{2}$ and the entire sum becomes the following:

$$\begin{aligned}\text{sum } &= 14(6000) + 24\left(\frac{14 \times 15}{2}\right)\\ &= 84000 + 12 \times 14 \times 15\end{aligned}$$

Using difference of squares on 12×14, we can see that this equals $(13 + 1)(13 - 1) = 13^2 - 1 = 169 - 1 = 168$.

$$84000 + 168 \times 15.$$

It might be easier to make the product into $168 \times \frac{30}{2}$, which equals 2520. (Or, it might not! It's up to you.) Finally, the total sum equals $84000 + 2520 = \boxed{86520}$.

15 Arnav, Bing and Chirag are playing a total of 30 card games, and bet $\$1$ on each game. The two losers pay $\$1$ to the winner of each game. If Bing ends up with a net positive total of money $\$21$ more than Chirag and Arnav ends up with a net $\$15$ more than Chirag, how many games did Arnav win?

✕	**Your Answer**	[no answer given]
	Correct Answer	11

Explanation

Let a equal the number of games win by Arnav, b equal the number of games won by Bing, and c equal the number of games won by Chirag. We saw in a previous question that we can write equations for the net total money Arnav, Bing, and Chirag get.

$$\begin{aligned} \text{Arnav's net money} &= 3a - 30 \\ \text{Bing's net money} &= 3b - 30 \\ \text{Chirag's net money} &= 3c - 30 \end{aligned}$$

These equations tell us that the difference in net total money between any two people equals 3 times the difference in the number of games that they won. For example, the net money difference between Arnav and Bing is $3a - 30 - (3b - 30) = 3(a - b)$. Using the given information, Bing makes $\$21$ more than Chirag, so he wins $\frac{21}{3} = 7$ more games than Bing. We can write $b = 7 + c$.

Similarly, Arnav makes $\$15$ more than Chirag, so he wins $\frac{15}{3} = 5$ more games than Chirag. We can write $a = 5 + c.$.

There are 30 games total, so we can write the following:

$$\begin{aligned} a + b + c &= 30 \\ (5 + c) + (7 + c) + c &= 30 \\ 12 + 3c &= 30 \\ 3c &= 18 \\ c &= 6. \end{aligned}$$

Arnav wins 5 more games than Chirag, so he wins $6 + 5 = \boxed{11}$ games.

16 In Expiiland, a person's age is determined by their name. Each letter has a value, and the age of a person is the product of the letters in his or her name. A group of Expiidians, wanting to know the value of each letter, visit the town elder for help. The elder agrees to release the letter values, but only if the group can tell him the current age of the town founder, Sylbia. What number should the group give the elder? Your answer should the number only, without commas or units.

The Group of Expiidians
Name: Bill, Age: 10
Name: Lila, Age: 20
Name: Sally, Age: 30
Name: Ali, Age: 40

Your Answer	[no answer given]
Correct Answer	2400

Explanation

We can set up equations for each name to solve for specific variables and batches of variables.

$$l = \frac{\text{lila}}{\text{ali}} = \frac{20}{40} = \frac{1}{2}$$

$$\text{say} = \frac{\text{sally}}{ll} = \frac{30}{\frac{1}{4}} = 120$$

$$\text{bi} = \frac{\text{bill}}{ll} = \frac{10}{\frac{1}{4}} = 40$$

$$\text{sylbia} = \text{say} \times \text{bi} \times l = 120 \times 40 \times \frac{1}{2} = 2400$$

Therefore, the group should tell the elder that Sylbia is $\boxed{2400}$ years old.

17 In one beehive, honey is produced at a rate of 2 teaspoons per honeybee. In a second hive, honey is produced at a rate of 3 teaspoons per honeybee. Suppose that over the course of a month, the honey produced both hives was the same. What was the average number of teaspoons of honey that one bee produced during this time? Your answer should be a simplified fraction in the form $\frac{a}{b}$, where a and b are integers.

1. / 2.

Your Answer

1. [no answer given]
2. [no answer given]

Correct Answer

1. 12
2. 5

Explanation

The clue here is that the honey produced in both of the hives is the same. Recall we learned that we could apply the idea of harmonic mean to a situation where equals quantities are created at different rates. We can make analogous quantities to the quantities of distance, speed and time from the lesson example.

$$\text{Distance (km)} \to \text{Honey (teaspoons)}$$

$$\text{Speed} \left(\frac{\text{km}}{\text{hr}}\right) \to \text{Honey Rate} \left(\frac{\text{teaspoons}}{\text{bee}}\right)$$

$$\text{Time (hours)} \to \text{Bees}$$

If this analogy is a little confusing, try to imagine that the more bees you have, the more honey they make. Some bees are more efficient at making honey, and some are less efficient (analogous to different speeds). Using this analogy, the two speeds are $2\frac{\text{teaspoons}}{\text{bee}}$ and $3\frac{\text{teaspoons}}{\text{bee}}$. Recall that the harmonic mean of two speeds x and y was the reciprocal of the average of the reciprocals of the rates, which can be simplified like so:

$$\frac{1}{\frac{\frac{1}{x}+\frac{1}{y}}{2}}$$

$$= \frac{2xy}{x+y}$$

Thus the average honey production rate is:

$$\frac{2xy}{x+y} = \frac{2(2)(3)}{2+3}\ \frac{\text{tsp}}{\text{bee}}$$

$$= \boxed{\frac{12}{5}}.$$

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 An experimental aircraft is outfitted with both ramjet engines and normal jet engines, but can only use one or the other, not both at the same time. The ramjets accelerate by $21\frac{\text{mph}}{\text{sec}}$, and the normal jets accelerate the plane by $14\frac{\text{mph}}{\text{sec}}$. Suppose a plane's speed increase from ramjets is equal to its speed increase from normal jets. What was the average acceleration (rate of speed increase)? Your answer should be a decimal rounded to the nearest tenth, like 3.1 for π, for example.

✕	**Your Answer**	[no answer given]
	Correct Answer	16.8

Explanation

Notice that the speed increase from the ramjets is equal to the speed increase due to the normal jets, which is a clue here that we can apply the harmonic mean once again. What's the analogy between acceleration, speed increase, and time, and the quantities distance, speed, and time, in the example of harmonic mean, which we saw in the Day 5 lesson? In that lesson, distance was conserved, and the speed and time varied. Thus speed increase here is analogous to distance, and acceleration is analogous to speed. The average acceleration is equivalent to the harmonic mean of 21 and 14, which, using the formula derived in the lesson, is equal to the following:

$$\begin{aligned}\text{harmonic mean of 21 and 14} &= \frac{2(14)(21)}{14+21}\\ &= \frac{2\times 2\times 7^2\times 3}{5\times 7}\\ &= \frac{12\times 7}{5}\\ &= \frac{84}{5}\\ &= \boxed{16.8}\frac{\text{mph}}{\text{sec}}.\end{aligned}$$

19 Without using a calculator, what are the first 3 digits of the product of 234523452345 and 428571428571?

Your Answer	[no answer given]
Correct Answer	100

Explanation

The repeating sequence of 6 digits in the second number of the product is the same as that of $\frac{3}{7} = 0.\overline{428571}$. It so happens that $428571 \times \frac{7}{3} = 999999 = 10^6 - 1$ (and, likewise, $428571428571 \times \frac{7}{3} = 999999999999 = 10^{12} - 1$). To account for multiplying the second number by $\frac{7}{3}$, we must multiply the first by $\frac{3}{7}$.

Now, $2345 \times \frac{3}{7} = \frac{7035}{7} = 1005$, so $234523452345 \times \frac{3}{7} = 100510051005$.

Therefore, our original product is the same as

$$\begin{aligned} &100510051005 \times 999999999999 \\ &= 100510051005 \times 10^{12} - 100510051005 \times 1 \end{aligned}$$

The second term is insignificantly small compared to the first, so the first three digits of our product will be the same as the first three digits of $100510051005 \times 10^{12}$, or $\boxed{100.}$

20 Suppose you have the digits 123456789. Put either a "$+$" or "$-$" between each digit. The result is called "S." Now add up all the possible **positive** "S" that we can achieve. What is this sum?

Your Answer	[no answer given]
Correct Answer	529

Explanation

We saw in the mini-question to the video lesson that it's not possible to get an even sum. The question is whether we can get every odd sum, $1, 3, 5, 7, \ldots 43, 45$. This is equivalent to finding a subset of $1, 2, 3, 4, 5, 6, 7, 8, 9$ which sums to $22, 21, 20, 19, 18, \ldots, 3, 2, 1$, because subtracting twice of $22, 21, 20, 19, 18$, etc. from 45 will result in sums of $1, 3, 5, 7, 9, \ldots 41, 43$, respectively. We just need to show that it is possible to get sums of $22, 21, 20, 19, 18$, etc. from a subset of $1, 2, 3, 4, 5, 6, 7, 8, 9$.

It's obvious that it is possible to get a sum of $1, 2, 3, 4, 5, 6, 7, 8, 9$. The following is a list of ways to achieve any sum up to 45 :

$$\begin{aligned}
(0)+1,(0)+2,\ldots,(0)+9 &= 1 \text{ through } 9\\
(9)+1,(9)+2,\ldots,(9)+8 &= 10 \text{ through } 17\\
(9+8)+1,(9+8)+2,\ldots,(9+8)+7 &= 18 \text{ through } 24\\
(9+8+7)+1,(9+8+7)+2,\ldots,(9+8+7)+6 &= 25 \text{ through } 30\\
(9+8+7+6)+1,(9+8+7+6)+2,\ldots,(9+8+7+6)+5 &= 31 \text{ through } 35\\
(9+8+7+6+5)+1,(9+8+7+6+5)+2,\ldots,(9+8+7+6+5)+4 &= 36 \text{ through } 39\\
(9+8+7+6+5+4)+1,(9+8+7+6+5+4)+2,\ (9+8+7+6+5+4)+3 &= 40 \text{ through } 42\\
(9+8+7+6+5+4+3)+1,(9+8+7+6+5+4+3)+2 &= 43 \text{ through } 44\\
(9+8+7+6+5+4+3+2+1)=45 &
\end{aligned}$$

Therefore it's possible to get any odd number. The odds go from 1 to 45, which we can write as $1+2(n-1)$ for the nth odd. Therefore $45 = 1+2(n-1) \to n = 23$ and there are 23 odds. We saw in a different video lesson that the sum of the first n odds is equal to n^2, so the sum is $23^2 = \boxed{529}$.

Summary

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